TEXORA · LECTURE SHEET

Thermodynamics

The engineer's bookkeeping of energy: systems and properties, the three laws, enthalpy and entropy, the gas processes — and the vapour cycles that run the mill's boiler house, from Carnot's ceiling to a full Rankine calculation from the steam tables.

ME-3203 · THERMAL ENGINEERING AND FLUID MECHANICS · TOPIC 1 · LEVEL 3 TERM 2

1. Systems, properties and processes

A **thermodynamic system** is the region chosen for study; everything outside is the surroundings, and the boundary between them decides the bookkeeping. A **closed** system exchanges energy but not mass (gas in a cylinder); an **open** system (control volume) lets mass flow through (a boiler, a turbine, a dyeing machine's steam coil). State is fixed by **properties** — pressure p, volume V, temperature T, internal energy U — which depend only on the state, not on how it was reached. A **process** is a path between states; a **cycle** returns to its start, so every property ends where it began while heat and work generally do not cancel.

Two properties are manufactured for convenience. **Enthalpy** H = U + pV packages internal energy with flow work, so steady-flow devices and constant-pressure heating are handled in one term: at constant pressure ΔH equals the heat added. **Entropy** S, defined by dS = δQ_rev/T, measures the dispersal of energy; it is the currency of the second law (Section 3). Intensive properties (T, p) are independent of size; extensive ones (V, U, H, S) scale with mass and are quoted per kilogram (specific values v, u, h, s) throughout engineering practice.

2. The first law — energy is conserved

For a closed system, with W the work done *by* the system:

$$Q - W = \Delta U$$

Heat in is positive, work out is positive, and the difference is stored. The standard processes follow at once:

PROCESS	CONDITION	FIRST-LAW BOOKKEEPING
Constant pressure	p constant; W = pΔV	Q = ΔH — heat shows up as enthalpy rise
Constant volume	W = 0	Q = ΔU

Isothermal (ideal gas)	$\Delta U = 0$	$Q = W = mRT \ln(V_2/V_1)$
Adiabatic	$Q = 0$; $pV^{\gamma} = \text{const}$	$W = -\Delta U$ — work is paid from internal energy

WORKED EXAMPLE — HEATING AIR AT CONSTANT PRESSURE

1 kg of air ($c_p = 1.005$, $c_v = 0.718$ kJ/kg·K) is heated at constant pressure from 300 K to 400 K. Heat added $Q = mc_p\Delta T = 1 \times 1.005 \times 100 =$ **100.5 kJ**. Internal energy rise $\Delta U = mc_v\Delta T =$ **71.8 kJ**. Boundary work $W = Q - \Delta U =$ **28.7 kJ** — exactly $mR\Delta T = 0.287 \times 100$, as it must be: at constant pressure the gas spends part of the heat pushing its boundary out.

For steady-flow equipment (boiler, turbine, compressor, nozzle) the first law per kilogram reads $q - w = \Delta h + \Delta(ke) + \Delta(pe)$, which collapses device by device: in a boiler $w = 0$ and $q = h_2 - h_1$; in a turbine $q \approx 0$ and $w = h_1 - h_2$; across a throttle valve h stays constant — the Joule–Thomson expansion used in refrigeration and steam traps.

3. The second law and Carnot’s ceiling

The first law permits a ship to sail by cooling the sea; the second forbids it. **Kelvin:** no cyclic engine can convert heat from a single reservoir wholly into work — some heat must be rejected to a colder sink. **Clausius:** heat never flows cold-to-hot unaided. The quantitative consequence is **entropy**: in any real process the entropy of the universe increases, and that increase prices every irreversibility — friction, throttling, heat transfer across a finite temperature gap.

The most efficient cycle possible between two reservoirs is the reversible **Carnot cycle** (two isothermals joined by two adiabatics):

$$\eta_{Carnot} = 1 - T_{cold}/T_{hot} \quad \text{(kelvin)}$$

WORKED EXAMPLE — CARNOT CEILING FOR A MILL POWER PLANT

Steam is raised at 400 °C (673 K) and the condenser runs at 40 °C (313 K). $\eta_{Carnot} = 1 - 313/673 =$ **53.5%**. No engine between these temperatures — however perfect — can beat 53.5%. The practical Rankine plant below reaches about 35%, the gap being the price of irreversible heat transfer in the boiler, turbine losses and the sensible heating of feedwater. Carnot is a ceiling, not a design.

4. The Rankine cycle — the working steam plant

Real steam plants follow the **Rankine cycle**: pump the condensate up to boiler pressure (1 → 2 in liquid, cheap work), boil and superheat at constant pressure (2 → 3), expand through the turbine (3 → 4, roughly isentropic), condense at constant pressure (4 → 1). On T–s coordinates the cycle is a closed loop whose width measures net work and whose heat input is the area under the boiling line.

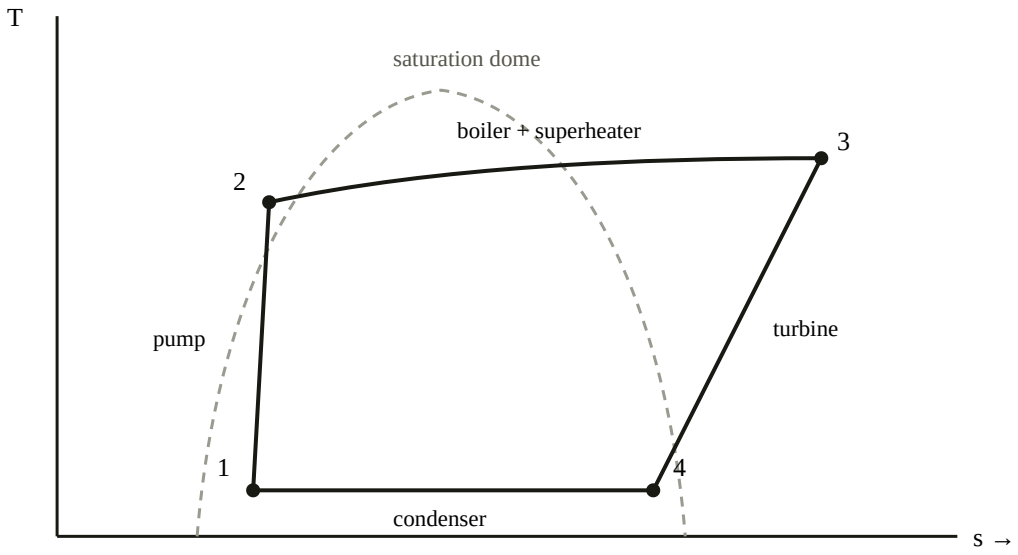


Figure 1 — The Rankine cycle on T–s coordinates. State 3 is superheated steam; 4 is wet mixture in the condenser. The dashed curve is the saturation dome.

WORKED EXAMPLE — RANKINE EFFICIENCY FROM THE STEAM TABLES

Steam at 30 bar, 350 °C: $h_3 = 3115.3$ kJ/kg, $s_3 = 6.7450$ kJ/kg·K. Condenser at 0.05 bar: h_f = 137.8, h_fg = 2423.1 kJ/kg; s_f = 0.4764, s_g = 8.3960 kJ/kg·K; v_f = 0.001005 m³/kg.

Isentropic expansion: $x_4 = (6.7450 - 0.4764)/(8.3960 - 0.4764) = 0.792$, so $h_4 = 137.8 + 0.792 \times 2423.1 =$ **2055.7 kJ/kg**. Turbine work = 3115.3 − 2055.7 = **1059.6 kJ/kg**.

Pump work = v_f × Δp = 0.001005 × (3000 − 5) = **3.0 kJ/kg** (note how small — liquids are nearly incompressible), so $h_2 = 140.8$ kJ/kg and net work = 1056.6 kJ/kg.

Heat supplied = 3115.3 − 140.8 = 2974.5 kJ/kg. η = 1056.6/2974.5 = **35.5%**; specific steam consumption = 3600/1056.6 = **3.41 kg/kWh**. Raising pressure or superheat temperature lifts efficiency but the exhaust dryness fraction (here 0.79) must not fall so low that water droplets erode the turbine blades — the classic design compromise.

Improvements on the basic cycle — **reheat** (expand in two stages with a second pass through the boiler) and **regenerative feed heating** (bleed steam to warm the feedwater) — push real plants toward 40%+ and dominate utility practice; in a textile mill the same boiler more often feeds process steam to dyeing, drying and stenters, with electricity bought in, but the cycle arithmetic is identical.

5. Key points

- ΔU = Q − W (closed system); H = U + pV packages flow work; steady-flow devices each collapse the energy equation to one or two terms.
- Entropy prices irreversibility; the second law forbids single-reservoir engines and free cold-to-hot heat flow.
- η_Carnot = 1 − T_c/T_h is an untouchable ceiling (53.5% between 400 °C and 40 °C).
- Rankine: pump (small work, ≈ vΔp) → boiler/superheater → turbine → condenser; the 30 bar/350 °C example gives 35.5% and 3.41 kg/kWh.
- Watch exhaust dryness: higher efficiency paths wet the turbine exhaust and erode blades; reheat and feed heating are the standard remedies.

6. Practice

Answer in your own words; brief model answers follow.

1. A gas in a cylinder absorbs 800 kJ of heat and does 300 kJ of work. Find ΔU. In a second process returning it to the start state, 200 kJ of work is done *on* the gas; how much heat is rejected?
2. Find the Carnot efficiency between 500 °C and 30 °C, and the maximum work from 10 MJ of heat supplied at the hot side.
3. 0.5 kg of steam at constant pressure receives heat raising its enthalpy by 400 kJ/kg while its internal energy rises by 350 kJ/kg. Find the boundary/flow work equivalent per kilogram, and total heat added.
4. In the Rankine example, what happens to efficiency and to exhaust dryness if condenser pressure falls to 0.03 bar (qualitative, with the mechanism)?
5. A refrigerator keeps a cold store at 5 °C in a 35 °C plant room. Find its Carnot COP and interpret the number.
6. Why is pump work in a Rankine cycle so much smaller than turbine work, although both handle the same mass flow between the same pressures?

Answer key

1. ΔU = 800 − 300 = +500 kJ. Round the cycle ΔU_total = 0, so the return leg has ΔU = −500 kJ; with W = −200 kJ: −500 = Q − (−200) → Q = −700 kJ: 700 kJ of heat is rejected.
2. T_hot = 773 K, T_cold = 303 K. η = 1 − 303/773 = 60.8%. Max work = 0.608 × 10 = 6.08 MJ; at least 3.92 MJ must be rejected.
3. Δh = Δu + Δ(pv): work equivalent = 400 − 350 = 50 kJ/kg. Total heat = 0.5 × 400 = 200 kJ (at constant pressure, Q = ΔH).

4. Efficiency rises (the cycle’s mean rejection temperature falls, widening the loop’s useful area); dryness falls — expansion ends deeper inside the dome, so more liquid droplets strike the last-stage blades. This is why deep vacuum is paired with reheat or higher superheat.
5. COP_Carnot = T_c/(T_h − T_c) = 278/(308 − 278) = 9.3. Each kilojoule of work can in principle pump 9.3 kJ of heat out of the store — refrigeration moves heat, it does not “make cold”, and the COP above 1 is no free lunch because the rejected heat Q_h = Q_c + W grows equally.
6. The pump compresses liquid water (v ≈ 0.001 m³/kg) while the turbine expands steam (v up to ~30 m³/kg at condenser pressure). Work scales with ∫v dp — three to four orders of magnitude more volume change on the expansion side, hence pump work ≈ 3 kJ/kg against turbine work ≈ 1060 kJ/kg.

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